

```
Processando config 30, seed 3, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config30_seed3.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells =  $2^L$  = 274877906944
Expected number per cell =  $1 / 32768$ 
EColl =  $n^2 / (2k)$  = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density =  $n / k = 1 / 32768.00$ 
Expected number of collisions =  $\mu = 128.00$ 
```

```
-----
Results of CollisionOver test:
```

```
POISSON approximation :
Expected number of collisions =  $N \cdot \mu$  : 256.00
Observed number of collisions : 276
p-value of test : 0.11
```

```
-----
Total number of cells containing j balls
```

```
j = 0 : 549739036948
j = 1 : 16776664
j = 2 : 276
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

```
-----
CPU time used : 00:00:04.46
```

```
Generator state:
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config30_seed3.bin
```

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 18.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.05

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 8.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.07

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : -1.97  
p-value of test : 0.98

-----  
CPU time used : 00:00:00.01

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 4.09  
p-value of test : 0.77

-----  
Normal statistic for number of jumps : -53.07  
p-value of test : 1 - eps1 \*\*\*\*\*

-----  
CPU time used : 00:00:00.05

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : 0.040  
p-value of test : 0.48

-----  
CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : 0.30  
p-value of test : 0.38

-----

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.030  
p-value of test : 0.38

Kolmogorov-Smirnov- statistic = D- : 0.021  
p-value of test : 0.63

Anderson-Darling statistic = A2 : 0.87  
p-value of test : 0.43

-----

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 0.46  
p-value of test : 0.93

-----  
Global longest run of 1 : 22.00  
p-value of test : 0.63

-----  
CPU time used : 00:00:00.05

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 102.85  
p-value of test : 0.25

-----  
CPU time used : 00:00:00.01

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 22.58  
p-value of test : 0.43  
-----

CPU time used : 00:00:00.00

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 1.57  
p-value of test : 0.06  
-----

CPU time used : 00:00:00.00

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : -0.093  
p-value of test : 0.54  
-----

CPU time used : 00:00:00.00

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----  
Normal statistic : -0.74  
p-value of test : 0.77

-----  
CPU time used : 00:00:00.01

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 216.68  
p-value of test : 0.02

-----  
CPU time used : 00:00:00.00

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 318.12  
p-value of test : 0.29

-----  
CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 473.11  
p-value of test : 0.69

-----  
CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :    1.19
p-value of test          :    0.12
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config30_seed3.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -2.58
p-value of test          :    0.9950
-----
```

```
CPU time used            : 00:00:00.00
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config30_seed3.bin
```

```
sstring_Run test:
```

```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355446
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 35.85
p-value of test : 0.48
-----
```

```
-----
Total number of bits: 13414816
```

```
Normal statistic for number of bits : -1.34
p-value of test : 0.91
```

-----  
CPU time used : 00:00:00.04

Generator state:  
299691904 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 6.63  
p-value of test : 0.08

-----  
CPU time used : 00:00:00.07

Generator state:  
316469120 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 1056.73  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.03

Generator state:  
333160320 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 55.40 |
| p-value of test              | : | 0.08  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 35.92 |
| p-value of test              | : | 0.73  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 64    |
| ChiSquare statistic          | : | 60.08 |
| p-value of test              | : | 0.62  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 36    |
| ChiSquare statistic          | : | 35.94 |
| p-value of test              | : | 0.47  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 22    |
| ChiSquare statistic          | : | 20.05 |
| p-value of test              | : | 0.58  |

-----  
CPU time used : 00:00:00.04

Generator state:  
349937536 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 94     |
| ChiSquare statistic          | : | 114.63 |
| p-value of test              | : | 0.07   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 94     |
| ChiSquare statistic          | : | 125.07 |
| p-value of test              | : | 0.02   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 512    |
| ChiSquare statistic          | : | 528.71 |
| p-value of test              | : | 0.30   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 90    |
| ChiSquare statistic          | : | 73.12 |
| p-value of test              | : | 0.90  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 50    |
| ChiSquare statistic          | : | 51.37 |
| p-value of test              | : | 0.42  |

-----  
CPU time used : 00:00:00.04

Generator state:

366714752 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config30\_seed3.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 112.65 |
| p-value of test              | : | 0.86   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 137.03 |
| p-value of test              | : | 0.34   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 151.29 |
| p-value of test              | : | 0.70   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 141.38 |
| p-value of test              | : | 0.23   |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 89    |
| ChiSquare statistic          | : | 92.31 |
| p-value of test              | : | 0.38  |

-----  
CPU time used : 00:00:00.03

Generator state:

383491552 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:05.94  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test                     | p-value  |
|--------------------------|----------|
| -----                    | -----    |
| 5 LinearComp             | 1 - eps1 |
| 22 MatrixRank, 320 x 320 | eps      |
| -----                    | -----    |

All other tests were passed